

Wildfire and Cloud-Forming Particles, Project Midway Report

A cumulative summary: models, trust checks, and the causal result

Motivation

Marine low clouds cool the planet by reflecting sunlight, and their brightness is governed largely by the number of cloud-condensation nuclei (CCN, the particles on which cloud droplets form). How strongly aerosols modify these clouds remains one of the largest uncertainties in estimates of radiative forcing, chiefly because aerosol and meteorology are entangled: the weather that carries particles to a site also alters clouds on its own. Episodic wildfire plumes offer a way in, perturbing an otherwise clean marine environment over days and hundreds of kilometres, a scale and duration relevant to climate.

Our question is the first link in that chain: does wildfire smoke reaching the Eastern North Atlantic (ENA) change CCN, and by how much? Wildfires cannot be randomized, so we turn to generative models: for each 10-day airmass trajectory we learn the *whole distribution* of CCN it could produce, then compare “with wildfire” against “without” at the same weather. Because aerosols persist for days, CCN on arrival reflects the history of the airmass rather than local conditions alone, which is why the trajectory, not the endpoint, is our unit of analysis. This report covers the models we used, whether we trust them, and what they say.

This project was motivated by Zhou et al.¹. That work addresses prediction; we address causal inference.

Data and split

Each sample is a 10-day back-trajectory timeseries: 241 hourly steps $\times$ 22 weather variables, with target $\log_{10}(\text{CCN})$ at arrival. We use the month-block (“paper”) split, which mirrors the motivating paper: six alternating months of 2022 are held out and the remainder used for training.

	Train	Test
Trajectories	57,646	3,220
Wildfire share	11.1%	12.7%
Independent episodes	$\sim 2,400$	~ 135

Two facts carry through the rest of the report: wildfire is a *small minority* of the data ($\sim 11\%$), and adjacent hours are $\sim 99.7\%$ identical, so tens of thousands of rows amount to only a few hundred *independent* episodes.

¹S. Zhou, D. Qi, H. Liu, et al., “A Lagrangian Time-Series Machine Learning Framework for Predicting Concentrations and Exploring Drivers of Atmospheric Aerosols: Model Development and Application to Cloud Condensation Nuclei in the Marine Boundary Layer,” AGU Fall Meeting 2024, abstract 1650614, <https://agu.confex.com/agu/agu24/meetingapp.cgi/Paper/1650614>; preprint: ESS Open Archive, doi:10.22541/essoar.177038612.23579844.

The models we used

Everything is an *engression* model: a sampler $\hat{Y} = g(X, \varepsilon)$ that takes the weather X plus random noise ε and returns one possible CCN; draw it many times and you trace out $P(Y | X)$. Training uses the energy-score (CRPS) loss, which rewards the whole predicted distribution, not just its mean. For the observed target y and two independent draws $\hat{Y} = g(X, \varepsilon)$, $\hat{Y}' = g(X, \varepsilon')$ from the model,

$$\mathcal{L} = \mathbb{E}|\hat{Y} - y| - \frac{1}{2} \mathbb{E}|\hat{Y} - \hat{Y}'|.$$

The first term pulls samples toward the truth; the second rewards spread, so the model cannot collapse to a single point. In practice we minimize its unbiased estimate over m draws $\hat{y}_1, \dots, \hat{y}_m$:

$$\hat{\mathcal{L}} = \frac{1}{m} \sum_{i=1}^m |\hat{y}_i - y| - \frac{1}{2m(m-1)} \sum_{i \neq j} |\hat{y}_i - \hat{y}_j|.$$

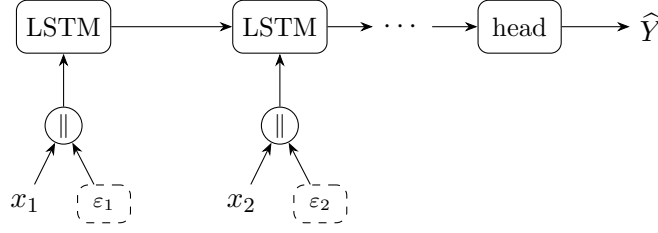
All three generators we carried into the causal analysis share one skeleton: an **LSTM reads the 241-step trajectory**, and a small head turns the result into a distribution over CCN. What separates them, and makes them stochastic, is **where the noise enters the recurrence**:

- **Per-timestep.** A fresh, independent 96-dim noise vector is *concatenated onto the 22 weather inputs at every one of the 241 steps* (so the LSTM input widens $22 \rightarrow 118$). Because the noise is redrawn at every step, this variant resists collapse most effectively of the three.
- **Global-latent.** The *same* wiring, except a single 96-dim code ε is drawn *once per trajectory* and concatenated to every step (the same ε all the way through), giving one shared “latent code” for the whole sequence.
- **Recurrent.** The exception: instead of concatenating noise to the inputs, it uses an **LSTMCell** and *adds* noise directly to the hidden state after each step, $h_t \leftarrow h_t + \text{softplus}(s) \varepsilon_t$ (with a learned per-unit scale s). Because the state carries forward, the randomness **accumulates** through the sequence.

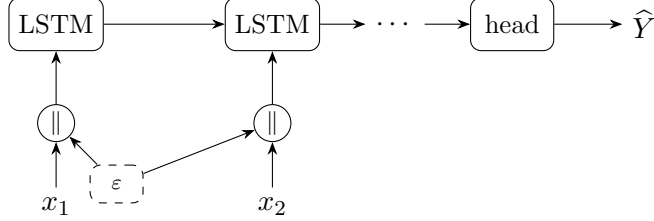
Generator	Noise location	Drawn	Randomness
Per-timestep	concat. to inputs, each step	fresh each step	independent per step
Global-latent	concat. to inputs, each step	once per trajectory	one shared code
Recurrent	added to hidden state, each step	fresh each step	accumulates over time

All three: noise dim 96, energy-score loss. Swept configs used downstream: per-timestep 5×128 , global-latent 3×128 , recurrent hidden 192.

(a) **Per-timestep** fresh ε_t concatenated to the inputs, every step



(b) **Global-latent** one ε drawn per trajectory, concatenated to every step



(c) **Recurrent** fresh ε_t added to the hidden state, accumulating

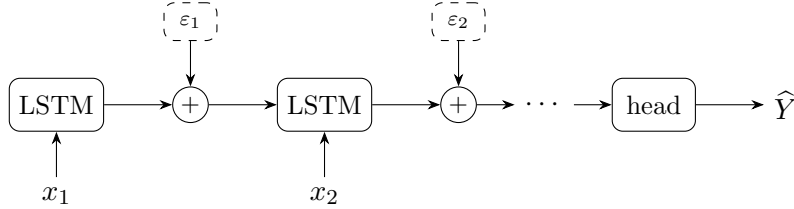


Figure 1: Where the noise enters each generator (two steps shown). Inputs x_t rise from below; the LSTM runs left to right into the head and output $\hat{Y}$; dashed nodes are noise. “||” concatenates noise to the inputs (a, b); “+” adds it to the hidden state (c). In (a) the noise is fresh each step; in (b) a single ε feeds every step; in (c) it accumulates along the chain.

Do we trust the models?

Two checks precede any causal claim.

Calibration (PIT). The *probability integral transform* rests on the fact behind inverse-transform sampling: if X has CDF F , then $F(X)$ is uniform on $[0, 1]$. Passing each observed CCN through its own predicted CDF should therefore yield **uniform** values whenever the predicted distribution is correct. In practice the PIT is the fraction of the model’s samples falling below the observed CCN, an empirical F . A well-calibrated model gives a **flat** histogram, whereas a **U-shape** indicates overconfidence, that is, intervals that are too narrow. Across the 48 sweep configurations (Figure 2) most histograms are mildly U-shaped and the best are nearly flat. All histograms and A^2 statistics below use the full held-out test set of 3,220 trajectories, with 400 predictive draws per trajectory and no subsampling.

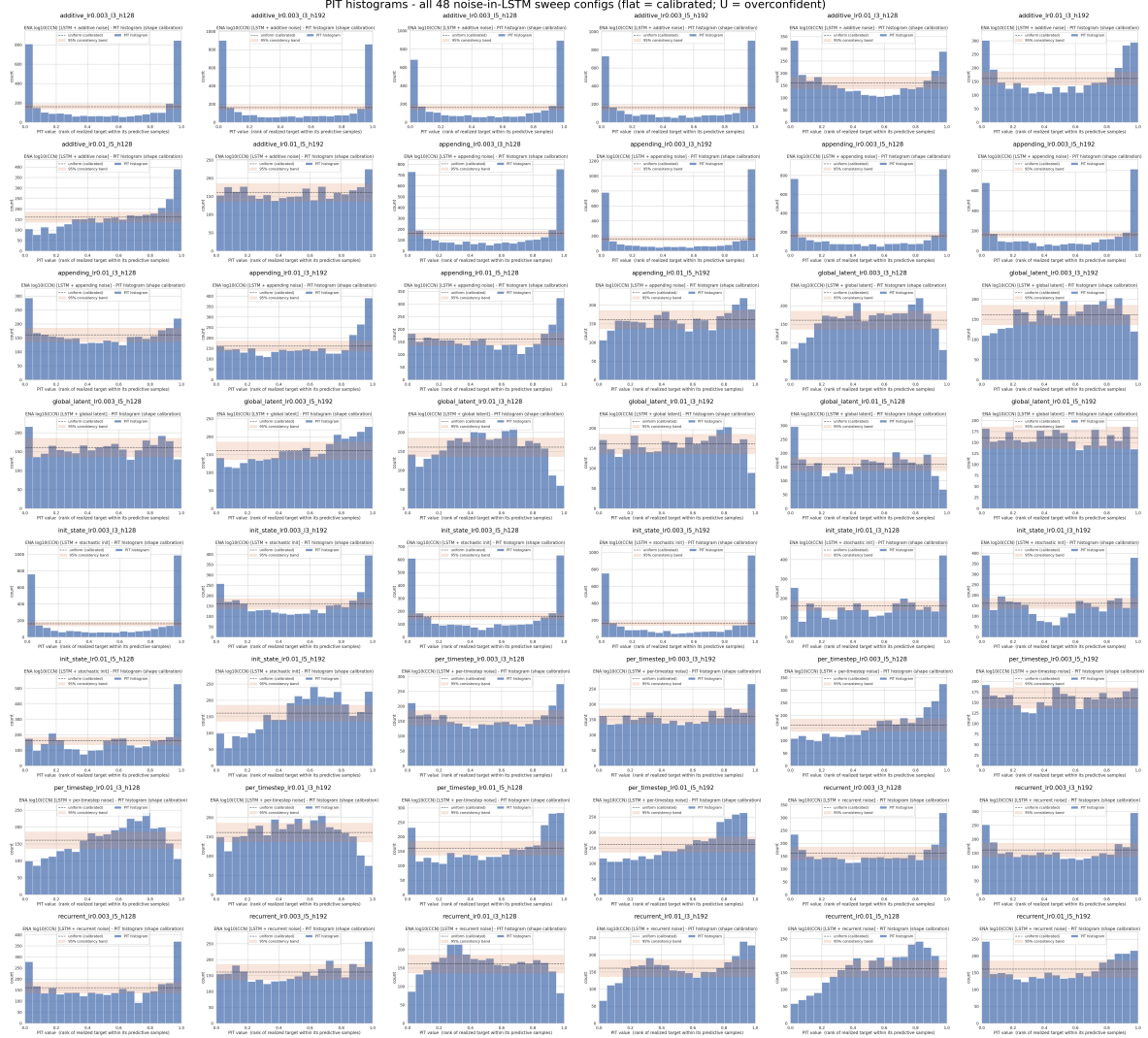


Figure 2: PIT histograms for all 48 configs. Flat = calibrated; U-shape = overconfident. Most are mildly U-shaped, the best near flat.

Quantifying uniformity. To move beyond visual inspection of the 48 histograms, we summarize each with the Anderson–Darling statistic A^2 , a scalar measure of a sample’s departure from uniformity. Under correct calibration A^2 has expectation near unity, and it grows as the histogram becomes more U-shaped. As an informal guide, $A^2 \lesssim 2.5$ is indistinguishable from uniform, values in the tens indicate a visible departure, and values in the hundreds or beyond indicate pronounced overconfidence. The table below reports, for each noise variant, the best, median, and worst A^2 over its eight swept configurations.

Noise variant	Best A^2	Median A^2	Worst A^2
Global-latent	1.0	23.8	56.4
Per-timestep	3.8	46.5	170.6
Additive	6.1	601.9	1252.6
Recurrent	16.3	37.6	128.4
Appending	20.0	482.9	1583.3
Stochastic-init	96.0	251.2	1661.6

Anderson–Darling A^2 against the uniform distribution, summarized over the eight configurations (learning rate $\times$ depth $\times$ width) of each variant and ordered by best A^2 . Because consecutive hourly trajectories overlap substantially, A^2 is best read as a relative ranking rather than an exact test statistic; only the values in the hundreds and above correspond to unambiguous miscalibration.

The global-latent generator is the best calibrated of the six: it attains the lowest score ($A^2 = 1.0$, effectively uniform) and never exceeds 56 anywhere in the sweep. The per-timestep and recurrent variants are also consistently well calibrated, which supports their use as the downstream generators. The remaining three variants can degrade to severe overconfidence, with A^2 reaching the thousands; these failures arise exclusively at the smaller learning rate (0.003), while the same architectures trained at the larger rate remain well calibrated.

Is the test weather in-distribution? The 2022 test weather differs significantly from the training weather: a classifier separates the two at AUC 0.79, and an MMD permutation test gives $p = 0.0008$, driven mostly by carbon monoxide, a smoke tracer. Different, however, is not the same as unseen. Test trajectories still lie *inside* the training region (nearest-neighbour distance ratio 1.04, with only 7.4% beyond the training 95th percentile; Figure 3), so the overlap the causal comparison requires does hold.

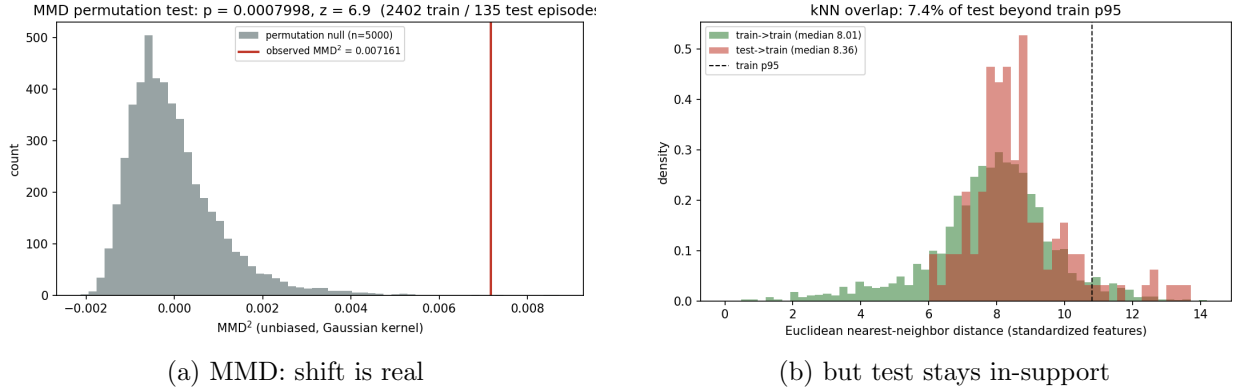


Figure 3: (a) Observed MMD (red) sits far outside the “no difference” null (grey): the train/test shift is statistically real. (b) Yet test→train distances (red) track train→train (green): the test weather stays inside the training region.

What relates to CCN?

No single variable drives CCN (Figure 4). The strongest correlate is downwelling solar radiation (SWGDN, +0.33), followed by cloud water, low cloud, and wind speed (all ≈ -0.25); particles are more

abundant on sunny, calm, low-cloud days. Pearson and Spearman correlations are nearly identical, so the marginal relationships are close to linear and the nonlinearity lies in how the variables combine. CO deserves separate mention: although it separates train from test more sharply than any other variable, it barely correlates with CCN on its own. It marks *when* smoky air arrives rather than how much CCN is present.

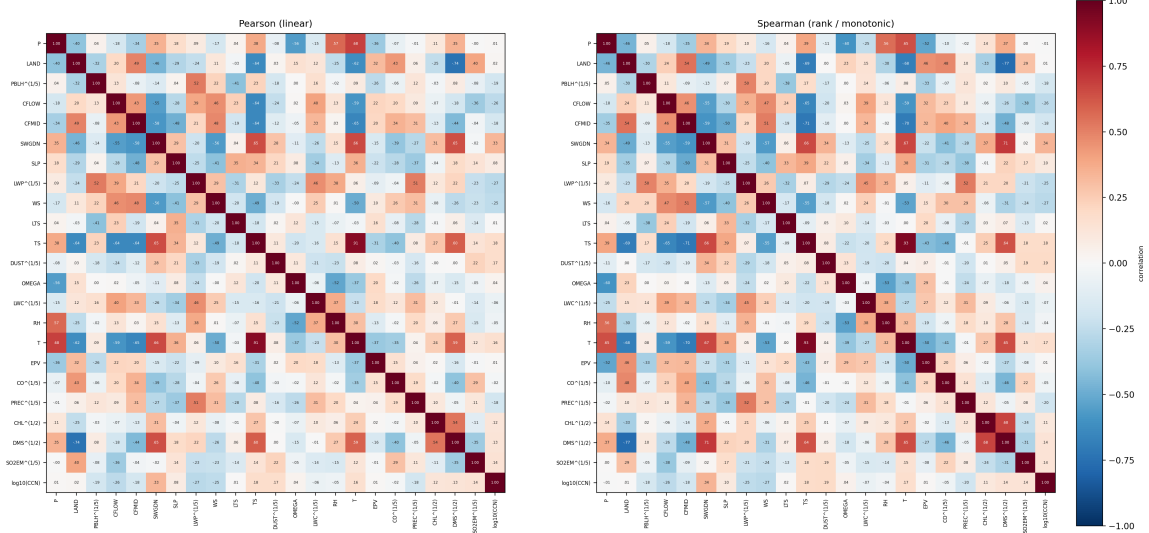


Figure 4: Correlation of all 23 variables (per-sample trajectory means, 20,000 samples). Left = Pearson (linear), right = Spearman (rank). Their near-identity means the marginal relationships are close to monotone-linear.

Covariates sensitive to wildfire

The covariates divide into two groups: those negligibly affected by smoke (*wildfire-robust*) and those that shift measurably during wildfire periods (*wildfire-sensitive*). The distinction matters twice over. Sensitive variables are potential confounders, and their unperturbed values during wildfire periods are never observed, so any counterfactual conditioning on them must model those values rather than read them off. We therefore examine all 22, comparing for each the distribution inside wildfire periods (BB_criterion1, the same treatment definition used by the T-learner; 6,786 trajectories) against outside them, taking the per-trajectory mean as the sample unit, consistent with Figure 4.

Season confounds the naive comparison. Wildfire periods are strongly seasonal: 84% of wildfire trajectories arrive between June and September, and January and December contain none at all. The wildfire-free group is therefore weighted toward winter, so any covariate with an annual cycle differs between the two groups whether or not smoke affects it. Comparing the raw distributions largely recovers the seasonal cycle. To break that confound we repeat every comparison against a *month-matched* subsample of wildfire-free trajectories, drawn to reproduce the wildfire group’s calendar-month distribution (17,337 trajectories); season can then no longer explain a difference.

Three measures quantify each difference: the two-sample Anderson–Darling statistic A^2 (scale-free and tail-sensitive), the Wasserstein-1 distance, and the energy distance. The distances carry the covariate’s units, so the table reports them standardized by the pooled standard deviation $\hat{\sigma}$: $W_1/\hat{\sigma}$

is the average displacement between the two distributions measured in standard deviations, and the energy distance is squared before standardizing so that the ratio is dimensionless. Once standardized the three agree closely (pairwise Spearman correlations of about 0.99), so the ordering does not depend on the choice of statistic. Because the samples are large and temporally autocorrelated, nearly every test is formally significant; the statistics are best read as a ranking of shift severity rather than as tests.

Covariate	A^2		$W_1/\hat{\sigma}$	$E^2/\hat{\sigma}$	Wildfire quantiles (%)					Matched wildfire-free (%)				
	raw	matched			5	25	50	75	95	5	25	50	75	95
CO ^{1/5}	331	2037	0.787	0.369	0.037	0.039	0.040	0.041	0.044	0.037	0.038	0.039	0.040	0.042
LAND	59	768	0.440	0.134	0	0.0041	0.137	0.290	0.485	0	0	0.029	0.178	0.407
PBLH ^{1/5}	825	593	0.417	0.113	3.17	3.35	3.50	3.65	3.85	3.22	3.46	3.61	3.74	3.91
LWP ^{1/5}	387	491	0.420	0.097	0.276	0.397	0.478	0.541	0.612	0.363	0.452	0.507	0.558	0.623
DMS ^{1/2}	1038	451	0.330	0.075	0.671	0.968	1.21	1.41	2.02	0.724	1.15	1.30	1.59	2.11
SLP ($\times 10^5$)	305	350	0.303	0.066	1.010	1.014	1.018	1.021	1.026	1.010	1.015	1.019	1.023	1.028
P	99	253	0.230	0.042	746	846	905	956	1002	754	866	927	981	1009
EPV ($\times 10^{-7}$)	87	247	0.192	0.033	-0.04	2.66	4.18	5.84	9.55	-0.60	1.56	3.47	5.42	9.32
SO2EM ^{1/5}	77	196	0.228	0.031	0.00053	0.001	0.0015	0.0022	0.0036	0.00038	0.00081	0.0013	0.002	0.0031
RH	39	156	0.214	0.031	0.541	0.665	0.747	0.812	0.876	0.559	0.702	0.770	0.825	0.885
CHL ^{1/2}	1195	145	0.270	0.024	0.259	0.346	0.441	0.570	0.951	0.234	0.315	0.426	0.536	0.724
CFLOW	1081	143	0.200	0.027	0.064	0.162	0.273	0.394	0.581	0.080	0.196	0.313	0.430	0.597
DUST ^{1/5}	8	130	0.176	0.020	0.0091	0.010	0.012	0.013	0.015	0.0086	0.0099	0.011	0.013	0.015
OMEGA	0	108	0.192	0.021	-0.013	0.0018	0.015	0.031	0.051	-0.011	-0.0003	0.012	0.025	0.047
TS	2107	80	0.160	0.014	279	285	290	294	298	276	284	289	295	299
PREC ^{1/5}	21	76	0.134	0.013	0.035	0.087	0.136	0.195	0.285	0.026	0.073	0.124	0.184	0.284
CFMID	1076	67	0.118	0.008	0.011	0.046	0.086	0.141	0.232	0.0041	0.036	0.085	0.149	0.261
WS	1515	67	0.140	0.013	3.68	4.61	5.31	6.13	7.33	3.67	4.70	5.49	6.35	7.70
T	1757	56	0.130	0.010	269	279	284	290	296	266	278	285	291	297
SWGDN	2199	45	0.101	0.006	112	185	231	268	309	99.1	179	231	277	320
LWC ^{1/5}	13	33	0.073	0.004	0.0054	0.026	0.043	0.062	0.086	0.0012	0.025	0.044	0.063	0.089
LTS	89	4	0.038	0.001	13.1	15.5	17.0	18.5	21.0	13.1	15.5	17.0	18.6	20.6

Per-covariate comparison of wildfire vs wildfire-free trajectories (per-trajectory means), ordered by the month-matched A^2 . The “raw” column compares against all 54,080 wildfire-free trajectories; every other column compares against the month-matched subsample, as do the quantiles shown. The distance columns are standardized: $W_1/\hat{\sigma}$ and $E^2/\hat{\sigma}$ divide the Wasserstein-1 distance and the squared energy distance by the covariate’s pooled standard deviation, so both are dimensionless and comparable across rows (A^2 is scale-free by construction). Quantiles are on each covariate’s stored (power-transformed) scale, with SLP and EPV scaled as indicated.

Matching on month reorders the table almost completely, which is the main lesson of this section. The four covariates that dominate the raw comparison are precisely those with the strongest annual cycles, and they collapse once season is held fixed: SWGDN falls from $A^2 = 2199$ to 45 (first place to twentieth), T from 1757 to 56, and TS and WS similarly. Their apparent sensitivity to wildfire was largely the summer climatology of the Azores. The ocean-biology terms CHL and DMS fall for the same reason.

What rises is more physically suggestive. CO, a long-range tracer of biomass burning, moves from eleventh to first with A^2 increasing sixfold, and LAND, the fraction of the trajectory spent over land, from seventeenth to second. Together these describe continental air that has passed over burning regions, which is what a smoke signature should look like. PBLH and LWP, both plausible responses to absorbing aerosol aloft, remain high after matching. Note also that CO achieves the largest matched A^2 in the table while its central quantiles stay nearly unchanged: its shift lives in the upper tail, in the smoke episodes themselves, where a tail-sensitive statistic sees what quantiles near the median cannot.

Even so, this table cannot by itself identify wildfire-sensitive variables, and the LAND result shows why. Smoke does not alter the fraction of a trajectory spent over land; rather, trajectories that cross

land are the ones that collect smoke. A month-matched difference still conflates “smoke changes this variable” with “smoke arrives in air masses that already differ in this variable.” Season is only the most obvious such confounder, not the only one. Separating the two requires exactly the machinery the next section applies to CCN: a model of what each variable would have been in the same air mass without the smoke.

Preliminary results: wildfire raises CCN

We use a **T-learner**: one model is trained on wildfire trajectories and another on wildfire-free ones, and *both* are then run on the same weather. The wildfire arm gives the observed “with smoke” CCN, the clean arm gives the counterfactual “what it would have been without,” and the gap between them is the wildfire effect. Both arms use the stochastic LSTM generators described above, and the entire T-learner is run once per architecture: per-timestep (5×128), global-latent (3×128), and recurrent (hidden 192). Each row below, and each effect plot, is therefore one complete two-arm T-learner.

Our primary estimate is the **ATE (average treatment effect)**, which averages over *all* test weather and so captures the typical effect of wildfire across conditions. **Wildfire raises CCN by roughly 20–30%** ($\approx +30$ to $+40$ particles cm^{-3}), with the whole distribution shifting upward and developing a heavier upper tail (Figure 5). All three generators agree:

Generator	ATE (all weather)	ATT (wildfire weather)
Per-timestep	+20%	+10%
Recurrent	+31%	+16%
Global-latent	+31%	+24%

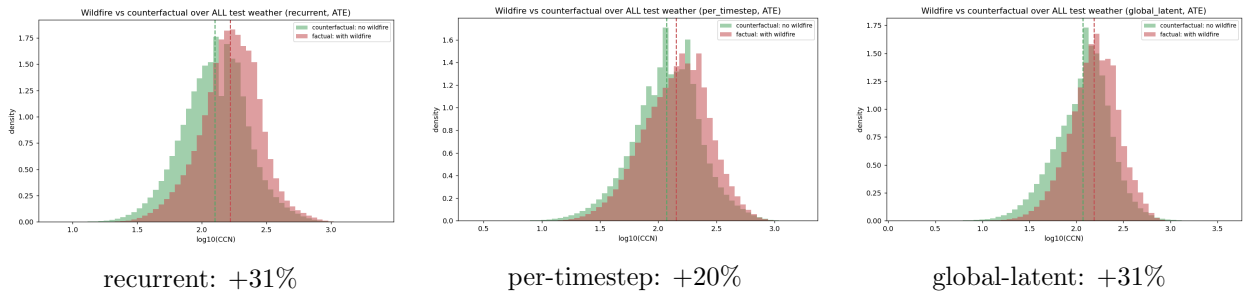


Figure 5: ATE view: CCN with wildfire (red) vs. the counterfactual without (green), both evaluated over *all* held-out test weather. Every generator shifts the distribution up (dashed = means) with a fatter high tail. Horizontal axis is a log scale.

ATE and ATT. The ATT (right column) restricts the effect to weather that actually occurred *during* wildfire periods, and is smaller (+10 to +24%). The ATE is larger because it also asks what wildfire would do on *clean*-weather days, which requires the wildfire arm to predict on conditions it rarely saw; part of the gap between the two therefore reflects extrapolation. The ATE is best read as the broad average effect and the ATT as the conservative, in-support estimate.

Bottom line. Across three stochastic generators, wildfire smoke raises cloud-forming particles at ENA by roughly **20–30%** on average (ATE), or a more conservative **10–24%** when restricted to wildfire conditions (ATT). The models are reasonably calibrated and the test weather remains in support, so the direction and approximate magnitude of the effect are well founded.

Caveats

The estimate rests on ignorability: the 22 weather variables must capture everything that differs between wildfire and clean air apart from the smoke itself. There are no formal error bars yet, and the number of independent wildfire episodes is far smaller than the number of observations, so +20% to +31% should be read as a range rather than as exact figures. Because the ATE averages over all conditions, it also depends in part on the data-poor wildfire arm extrapolating onto clean weather.

Next steps: which variables does smoke itself change?

Our comparison holds the weather fixed across the two arms, and that is what makes it causal. But it only works for weather that smoke does not affect. If smoke changes a variable and we hold that variable fixed anyway, we cancel out part of the very effect we are trying to measure, and the answer comes out too small.

So the covariates fall into two groups. Some are background conditions smoke cannot plausibly touch, and holding those fixed is exactly right. Others may respond to smoke, which shades the surface, warms the layer it sits in, and can alter cloud and humidity. Those are part of the effect, not background, and treating them as fixed would hide some of it. Sorting the 22 variables into these two groups is the next task.

The table in the previous section cannot do that sorting. A variable can differ during smoky periods because smoke changed it, but equally because wildfires happen in summer, because recent years bring both more fires and warmer oceans, or simply because smoky air arrives from over land. The first two explain most of the raw differences we saw, and the third cannot be removed by matching at all. The plan is therefore to ask the question directly, by estimating what each variable would have been in the same air mass had the smoke not been there:

- Train a model on smoke-free periods that predicts each variable from conditions smoke cannot affect, then apply it during smoky periods to obtain the “no smoke” version of that variable. Variables whose observed values depart from it are the ones smoke changes.
- Guard against fooling ourselves. Repeat the procedure on invented smoke periods chosen to resemble real ones in season and year: anything flagged there is a false alarm. A genuine effect should also grow with the amount of smoke, and appear only after the air meets the plume.
- Where a variable stays ambiguous, report the CCN result both with and without it, so the range is visible rather than hidden behind a judgement call.

Beyond that, CCN is a means to an end. The same two-arm machinery applies to the cloud properties themselves, such as droplet size, water content, and cloud cover.